MISSION BRIEFING · SCIENTIST/ENGINEER SC

EMERALD MISSION: ISRO CS NIGHT-BEFORE CHECKLIST

Wed 28 Oct, 2:00 pm → 1:00 am. 124 checkpoints, and every question pattern from the 2023 and 2025 papers is on board.

🚀 Launch: 29 Oct | Part A · +1 / −⅓ | Part B · no negative | 14 mission phases

Mission rules: tick a box only when you can solve that type of question *without looking*. **23·Q52** = 2023 paper, question 52 (in the app: Study › Past paper, or Quiz › Past papers · subject). Part A: guess only if you can rule out one option. Part B: answer all 15.

2:00 pm – 2:15 pm Warm-up & exam rules ___ / 4

- ☐ Read the **Exam blueprint** chapter: 80 CS Qs (+1 / −⅓) and 15 aptitude Qs (20 marks, no negative)
- ☐ Guess rule: eliminate one option → guess; can't eliminate anything → skip
- ☐ Pack now: admit card, photo ID, 2 black/blue ball pens, watch, water. Know the centre route & reporting time
- ☐ Remember: write AND bubble the booklet SET code on the OMR

2:15 pm – 3:30 pm Operating Systems (8 Qs each year) ___ / 16

- ☐ Process states: which transition is impossible (Waiting → Running) — **25·Q65**
- ☐ fork(): parent and child run concurrently / parent waits; 2^n processes; return values — **25·Q64**
- ☐ Schedulers: long-term = degree of multiprogramming, medium = swapping, short = fast dispatcher — **23·Q52**
- ☐ FCFS by hand with arrival times, avg TAT & WT (also SJF, SRTF, RR) — **25·Q67**
- ☐ Response / waiting / turnaround time definitions from a timeline diagram — **25·Q68**
- ☐ Banker's algorithm: Need = Max − Alloc, find a safe sequence — **25·Q69**
- ☐ Resource-allocation graph → wait-for graph; cycle = deadlock (single instance) — **25·Q66**
- ☐ Deadlock conditions + formula $R \geq n(k-1)+1$
- ☐ Page replacement by hand: Optimal, LRU, FIFO (+ Belady's anomaly) — **25·Q70**
- ☐ Race condition, critical section, semaphore for producer–consumer — **25·Q63, 23·Q53**
- ☐ Priority inversion & priority inheritance — **23·Q49**
- ☐ Linux = one-to-one thread model — **23·Q48**
- ☐ Trap = software interrupt; illegal memory access → trap to OS — **23·Q51, Q54**
- ☐ Segmentation: loader assigns segment numbers; best/first/next fit & locality — **23·Q50, Q9**
- ☐ Paging numericals: page-table size, EAT with TLB, disk SSTF/SCAN movement (likely variants)
- ☐ Do: Quiz → Past papers · OS, then 20 OS questions

3:30 pm – 4:30 pm Theory of Computation + Compiler (10 Qs each year) ___ / 12

- ☐ **S → aSb | T, T → bT | b ⇒ $a^m b^n$, m < n** — asked in BOTH years (**23·Q57, 25·Q80**). Also try S → aaSb | T, T → Tb | a — **23·Q60**
- ☐ Read a DFA/NFA diagram → regular expression (ab*, letter(letter|digit)*) — **23·Q55, Q7**
- ☐ Minimal DFA state count: 2nd-from-right = b → 4; all non-empty strings → 2 — **23·Q56, 25·Q79**
- ☐ Regex equivalences: (a*b)* + (b*a)* = (a+b)* — **25·Q78**
- ☐ Closure: regular ∩ CFL is CFL (not always regular); RE not closed under complement — **23·Q58, Q59**
- ☐ Strings generated by a small grammar ($a^n cb^n$) — **25·Q33**
- ☐ Ambiguous grammar: E → E+E | E×E | id gives exactly 2 trees; ambiguity ⇔ 2 LMD ⇔ 2 RMD — **23·Q8, 25·Q32**
- ☐ Compiler phase order; parser receives tokens; yacc builds parsers, lex builds scanners — **23·Q10, 25·Q30, Q31**
- ☐ LL(1) = Left-to-right, Leftmost, 1 lookahead; LR family power order — **23·Q11**
- ☐ Regex token matching e.g. 0x[0-9A-F]+ (case matters!) — **25·Q28**
- ☐ Token counting & which phase catches which error (likely variant)
- ☐ Do: Quiz → Past papers · TOC and · Compiler

4:30 pm – 4:45 pm Break ___ / 1

- [ ] Walk, water, snack — no phone scrolling

4:45 pm – 5:45 pm DBMS & SQL (5–7 Qs) ___ / 10

- [ ] Find the candidate key from FDs, then the highest normal form (partial dependency → not 2NF) — **25·Q48**
- [ ] Normal forms: 5NF = join dependency, 4NF = MVD — **23·Q34**
- [ ] ER → tables: 1:N FK on the N side; M:N needs its own table; count minimum tables — **25·Q44, Q46**
- [ ] select * from R, S = CROSS JOIN; trace an INSERT/SELECT puzzle ($Y = 2^x - 1$) — **23·Q31, Q30**
- [ ] DELETE with a sub-query on another table — **23·Q32**
- [ ] ACID: durability holds always, any failure — **23·Q33**
- [ ] Composite index (country, city, street): leftmost-prefix rule — **25·Q45**
- [ ] Views restrict sensitive columns; NoSQL = flexible schema — **25·Q43, Q42**
- [ ] Transactions under locking: deadlock & rollback scenarios; 2PL, isolation levels — **25·Q47**
- [ ] Do: Quiz → Past papers · DBMS, then 15 DBMS questions

5:45 pm – 6:45 pm Computer Networks (6 Qs each year) ___ / 12

- [ ] Sliding window: window (packets) for > X% utilisation = W·Tt / (Tt + RTT) — **23·Q12**
- [ ] CSMA/CD: Tt ≥ 2·Tp → min frame or data rate — **23·Q13**
- [ ] Header overhead % across layers (5 × 6 B on 270 B → 10%) — **23·Q14**
- [ ] Wi-Fi standards & speeds: b 11M, a/g 54M, ac 3.5G, ad 7G — **23·Q15**
- [ ] Error detection vs correction (parity/CRC detect; RS/LDPC correct); CRC detects double errors — **23·Q16, Q22**
- [ ] Hamming distance decoding: tie ⇒ cannot correct — **25·Q37**
- [ ] Subnet masks: are two IPs on the same /26? (router needed) — **25·Q41**
- [ ] TCP ideal window = bandwidth × RTT (in bytes) — **25·Q40**
- [ ] Blocking ICMP breaks Path MTU discovery — **25·Q39**
- [ ] Wrong default gateway = Layer 3 — **25·Q35**
- [ ] Ports table (22, 25, 53, 80, 443…), TCP vs UDP header sizes (likely variant)
- [ ] Do: Quiz → Past papers · Networks, then 20 CN questions

6:45 pm – 7:30 pm Digital Logic + Computer Organization (11 / 7 Qs) ___ / 13

- [ ] Gate identities: tied-input NAND/NOR = NOT; NAND of inverted inputs = OR; LOW only when inputs equal = XOR — **23·Q38, Q39, Q40**
- [ ] Circuit trace: NOR then NAND — which inputs give LOW — **25·Q55**
- [ ] MUX/DEMUX/encoder facts: 1:16 demux → 4 select lines; octal encoder → 3 OR gates; 1-of-32 from 3:8 → 4 — **23·Q35, 25·Q50**
- [ ] Full adder = 3 inputs, 2 outputs — **23·Q37**
- [ ] Shift-register delay = bits × clock period — **23·Q36**
- [ ] Ripple counter: pulses from state 1001 to 0000 (MOD-16 → 7) — **25·Q53**
- [ ] Gated D latch transparent when clock HIGH — **25·Q54**
- [ ] Binary fraction → decimal; 2's complement multiply in 8 bits — **25·Q49, Q51**
- [ ] K-map / SOP simplification (X′Y′Z′ + X′YZ′ = X′Z′) — **25·Q52**
- [ ] Vectored interrupt, superscalar, FIFO prefetch, micro-operation — **23·Q18–Q21**
- [ ] Reverse Polish (postfix) of A*B+C*D — **23·Q17**
- [ ] Pipeline cycles (k + n − 1), cache tag/index/offset, AMAT (likely variants)
- [ ] Do: Quiz → Past papers · Digital Logic and · COA

7:30 pm – 8:15 pm Dinner break ___ / 1

- [ ] Proper meal, short walk. Skim the Formula sheet once while eating if you like

8:15 pm – 9:00 pm Data Structures + Algorithms (12 / 7 Qs) ___ / 12

- [ ] Tree traversals: inorder from a drawn tree; preorder of BST → postorder — 25·Q1, 23·Q4
- [ ] BST inorder predecessor/successor — 23·Q26
- [ ] Diameter of a binary tree is O(n); heap sort O(n log n) — 23·Q3, Q29
- [ ] Substrings of length-m string = m(m+1)/2 — 23·Q46
- [ ] Identify the sort from intermediate arrays (insertion) — 25·Q7
- [ ] MST weight by Kruskal; MST has V − 1 edges — 23·Q1
- [ ] Match Floyd / Dijkstra / Kruskal / Bellman-Ford / multistage to problems — 23·Q2, 25·Q3
- [ ] Greedy = locally optimal choices; Huffman uses frequencies — 25·Q4, Q2
- [ ] Growth-rate ordering ($n \log n < n^{1.5} < n^{\log n} < 2^n$); matrix multiply O(mnp) — 23·Q24, Q45
- [ ] Horner's rule: minimum operations for a polynomial — 23·Q5
- [ ] Radix sort count (n·d·base) & priority-queue answer "any position" — 23·Q25, Q28
- [ ] Do: Quiz → Past papers · DS and · Algorithms

9:00 pm – 9:45 pm Information Security (5 → 9 Qs, rising) ___ / 9

- [ ] Symmetric keys for n people = n(n−1)/2 (16 → 120); n-bit key → 2^n keys — 25·Q8, Q13
- [ ] RSA: given p, q, d find e ($ed \equiv 1 \bmod \varphi$) — 25·Q38
- [ ] Message digest uses no key; X.509 = PKI certificates — 25·Q9, Q15
- [ ] X.800 services: non-repudiation, integrity, confidentiality, access control — 25·Q10
- [ ] IPsec ESP tunnel mode = confidentiality + integrity + hides private IPs — 25·Q36
- [ ] DNSSEC does not encrypt; firewall stops network scanning (not physical attacks); phishing — 25·Q12, Q14, Q11
- [ ] Ping of death = buffer overflow; CAPTCHA vs automated attacks; asymmetric crypto ≠ DoS protection — 23·Q71, Q75, Q74
- [ ] Match encryption / MAC cloning / ARP spoofing / fingerprinting; ingress / egress / NAT / stateful firewall — 23·Q72, Q73
- [ ] Do: Quiz → Past papers · Security, then 15 security questions

9:45 pm – 10:30 pm Discrete & Engineering Maths (7 Qs each year) ___ / 9

- [ ] Determinant = 0 when rows/columns are dependent (spot it, don't compute) — 23·Q41, 25·Q56
- [ ] Eigenvalues of a (block-)triangular matrix — 25·Q57
- [ ] Euler: V − E + F = 2; regular graph E = kV/2; "enclosed regions" excludes the outer face — 23·Q43, 25·Q5
- [ ] Graph facts: degree sum = 2E, girth, eccentricity/radius; number of simple graphs $2^{n(n-1)/2}$ — 23·Q42, Q6
- [ ] Logic: $(p \wedge q) \rightarrow r \equiv \neg p \vee \neg q \vee r$; truth table has 2^n rows — 25·Q59, 23·Q47
- [ ] Counting: no-two-consecutive subsets (Fibonacci, 377); $|A^n| = m^n$; words from distinct letters (GAGANYAAN → 24) — 25·Q60–Q62
- [ ] Probability of not drawing a colour; Poisson is discrete — 23·Q44, 25·Q86
- [ ] Triple integral with odd-function cancellation (likely variant of 25·Q58)
- [ ] Do: Quiz → Past papers · Maths

10:30 pm – 10:45 pm Break ___ / 1

- [ ] Stand up, water, eyes off the screen

10:45 pm – 11:30 pm Software Engg + AI/ML + Cloud & Big Data (≈ 20 easy marks) ___ / 11



- [ ] Cyclomatic complexity E − N + 2P; white-box; MC/DC; regression testing — **23·Q79, 25·Q71, 23·Q76, Q77**
- [ ] Alpha/beta = acceptance; CMM levels; risk-management order; V-model pairs — **25·Q72–Q74, Q76**
- [ ] Fan-in/fan-out statements; Function Point calculation (578 × 1.07) — **25·Q75, Q77**
- [ ] ER = static view — **23·Q80**
- [ ] AI agent types (no "Unity based"); decision trees do both; Q-learning = RL; hill climbing = search — **25·Q16–Q20**
- [ ] FOL connectives; hybrid Bayesian network; K-means = partitional — **23·Q61, Q62, Q65**
- [ ] Elasticity; Type-1 vs Type-2 hypervisor; para-virtualization modifies the guest OS — **25·Q21, Q22, 23·Q70**
- [ ] Private cloud, cloud bursting, virtualisation vs dual boot — **23·Q67–Q69**
- [ ] Big Data Vs (not Visualisation), HDFS purpose, Flink = streaming, NoSQL for unstructured, structured/semi/unstructured — **25·Q23–Q27**
- [ ] Cassandra = distributed; JMX = Java management API — **23·Q66, Q64**
- [ ] Do: Quiz → Past papers · SE, · AI and · Cloud

11:30 pm – 12:15 am Aptitude (Part B) + Programming ___ / 8



- [ ] Number series with ×n + n pattern (find the wrong term) — **23·Q92**
- [ ] Calendar: day of the week via odd days — **25·Q81**
- [ ] Syllogisms: either-or pair; "neither" — **25·Q88, Q89**
- [ ] Fact-based inference ("must be true") — **23·Q90, 25·Q82, Q83**
- [ ] Course of action / survey-bias / workplace judgement — **23·Q91, Q85, Q84**
- [ ] Mixtures, weights system, % increase, max rectangle area, cube volume — **25·Q85, Q87, 23·Q94, Q95, 25·Q84**
- [ ] Pointer arithmetic (p + 3 with 2-byte int); out-of-bounds ptr[4]; JVM is hardware independent — **23·Q27, 25·Q34, Q29**
- [ ] Do: Quiz → Past papers · Aptitude and · Programming

12:15 am – 12:45 am Final sweep of the past papers ___ / 3



- [ ] Quiz → Past papers (all) → finish any you haven't answered
- [ ] Quiz → Wrong: redo every past-paper question you missed today until correct
- [ ] Re-read the starred (★) questions

12:45 am – 1:00 am Formula sheet, then sleep ___ / 2



- [ ] One read of the Last-day formula sheet (tap ? on anything that feels shaky)
- [ ] Alarm set; admit card + ID + pens in the bag; lights out by 1:00



Mission complete. Recharge your ring — sleep — then go light up that paper.

